AI-Based Location-Adaptive Landslide Early Warning System Using Multi-Temporal Environmental Data

MS. J. JERALDIN AMALA PRIYA
Department of AI&DS
PSR Engineering College
Sivakasi, Tamil Nadu, India
jeraldinamalapriya@psr.edu.in

RAGAVI R
Department of AI&DS
PSR Engineering College
Sivakasi, Tamil Nadu, India
ragaviragavi9360@gmail.com

AATHI LAKSHMI N
Department of AI&DS
PSR Engineering College
Sivakasi, Tamil Nadu, India
muthuaathi957@gmail.com

NAVYA FRANCIS F
Department of AI&DS
PSR Engineering College
Sivakasi, Tamil Nadu, India
navyafrancis54@gmail.com

***Abstract*—Landslides are influenced by rainfall, soil moisture, topography, and other environmental factors. Existing AI-based systems mainly focus on landslide prediction and susceptibility mapping. However, prediction accuracy alone does not indicate how early a risk increase can be detected. This paper proposes a location-adaptive landslide early-warning framework that analyzes multi-temporal environmental data and tracks changes in predicted risk. Instead of considering individual predictions, the system identifies continuous risk escalation and generates an early-warning state. The framework is evaluated using lead-time, detection, false-alarm, and missed-event measures against historical landslide events. Thus, the proposed approach focuses on detecting increasing landslide risk before an event, rather than only predicting the event itself.**

***Index Terms*—Landslide prediction, artificial intelligence, machine learning, early warning system, temporal risk, risk escalation, environmental data, lead-time analysis, false alarms.**

I. Introduction

Landslides are among the most important geohazards affecting mountainous and hilly regions. A landslide can damage roads, bridges, buildings, agricultural land, communication infrastructure, and other critical assets. The consequences become more severe when the event occurs near populated areas or transportation corridors. Rainfall is a common trigger for shallow landslides, but the response of a slope to rainfall is not determined by rainfall alone. Soil moisture, antecedent wetness, slope geometry, land cover, geological conditions, and local hydrological processes can influence whether a rainfall episode results in slope failure. Consequently, landslide prediction is a difficult spatio-temporal problem rather than a simple threshold-classification task.

The development of artificial intelligence has created new opportunities for landslide analysis. Machine learning models can learn nonlinear relationships among conditioning factors and observed landslide occurrences. Recent reviews show that AI methods have become widely used for landslide detection, susceptibility assessment, and other forms of risk assessment [1]–[3]. Common models include logistic regression, support vector machines, random forests, gradient boosting, convolutional neural networks, recurrent neural networks, and hybrid approaches. These methods can achieve useful predictive performance when the input variables, labels, sampling strategy, and validation design are appropriate.

However, there is an important distinction between susceptibility prediction and operational early warning. A susceptibility map generally answers a spatial question: which locations are more susceptible to landslides? An early-warning system must additionally answer a temporal question: is the risk at a monitored location changing in a way that suggests an approaching event? This distinction motivates the present work.

A model that produces a risk probability for every time step may generate a sequence such as 0.20, 0.23, 0.28, 0.35, 0.45, and 0.60. Interpreting only the final value ignores the information contained in the trajectory. The sequence suggests that the estimated risk is increasing continuously. The proposed framework therefore treats the risk prediction as a time series and introduces a risk-escalation layer. The purpose of this layer is not to claim that every increasing sequence will result in a landslide. Instead, it provides a measurable warning state that can be evaluated retrospectively against observed events.

The importance of temporal environmental information is supported by recent research. For example, a 2025 study demonstrated that incorporating in-situ soil moisture with meteorological data can improve precipitation-based shallow landslide hazard prediction and emphasized the value of sequential data for capturing temporal dependencies [5]. An-

other 2026 study proposed a multi-source monitoring framework using unsupervised machine learning and reported that combining monitoring variables can help address difficulties associated with fixed thresholds and reduce false alarms [6]. These studies indicate that temporal and multi-source information is relevant to modern early-warning research. Therefore, the contribution of this work should not be framed as the first use of AI for landslide warning. Instead, the proposed research focuses on a unified evaluation workflow in which future risk prediction, temporal escalation detection, lead-time measurement, and warning reliability are evaluated together.

The proposed system uses historical observations to train an AI model. Historical rainfall, soil-moisture, terrain, and event information are transformed into temporal features. The model then estimates the probability of a future landslide condition from new observations. A second layer observes the sequence of predictions and identifies sustained escalation. If an escalation criterion is satisfied, the system generates a warning. The warning is compared with the timestamp or event window of an observed landslide. The time difference is used to calculate lead time. Warnings that do not correspond to an event within the defined evaluation window are counted as false alarms, while events that occur without a preceding warning are counted as missed events.

The central research question is therefore formulated as follows:

> Can an AI-based landslide-risk model, combined with temporal risk-escalation analysis, provide an early-warning assessment that is evaluated not only by predictive performance but also by warning timeliness and reliability?

The main objectives are:

1) to construct a temporally aligned landslide dataset from historical environmental and event records;
2) to train and evaluate an AI model for future landslide-risk estimation;
3) to monitor the predicted risk trajectory and identify sustained risk escalation;
4) to quantify the time between escalation detection and observed landslide occurrence;
5) to evaluate false alarms and missed events together with conventional classification metrics; and
6) to establish a reproducible experimental framework that can be extended to additional monitoring sources.

The remainder of the paper is organized as follows. Section II reviews relevant landslide AI research. Section III defines the research gap and contribution. Section IV presents the proposed framework. Section V describes data requirements and preprocessing. Section VI discusses model development and implementation. Section VII describes the experimental design and evaluation metrics. Section VIII presents the planned results and discussion structure without fabricating experimental values. Section IX discusses limitations and future work. Section X concludes the study.

II. Background and Related Work

A. Landslide Susceptibility Assessment

Landslide susceptibility assessment estimates the relative likelihood of landslide occurrence across a geographical area. Traditional approaches use expert knowledge, statistical relationships, physically based models, or heuristic weighting. Machine learning methods have become increasingly common because they can represent nonlinear relationships among conditioning factors. A 2024 review of machine-learning-based susceptibility modeling emphasized issues such as class imbalance, sampling strategy, uncertainty, and the selection of conditioning variables [4]. Another broad review covering AI-based landslide risk assessment categorized major applications into detection, susceptibility assessment, and displacement prediction [1].

Random forests are frequently used because they can handle nonlinear relationships, mixed variables, and interactions among features. Support vector machines can perform well in high-dimensional spaces, while gradient boosting methods can capture complex decision boundaries. Deep learning methods can learn hierarchical representations but generally require more data and careful regularization. Reviews of recent studies also emphasize that model performance depends strongly on the quality and design of the evaluation index system [2].

Susceptibility studies are highly relevant to the proposed work because they establish the value of environmental and terrain conditioning factors. However, a susceptibility value is not necessarily an event-time warning. A location can be susceptible but remain stable for a long period. The proposed work therefore treats susceptibility-related variables as inputs to a temporal prediction problem rather than as the final output.

B. Rainfall-Based Early Warning

Rainfall thresholds have traditionally been used for landslide early warning. Such systems can use rainfall intensity, duration, cumulative rainfall, or intensity-duration relationships. Their main advantage is simplicity and interpretability. However, a rainfall-only threshold can have difficulty representing the antecedent moisture state of a slope. Two locations receiving the same rainfall may respond differently because their soil, vegetation, drainage, and prior moisture conditions differ.

Recent data-driven research has investigated the inclusion of soil moisture and meteorological time series. Halter et al. showed that combining in-situ soil moisture measurements with meteorological data can improve shallow landslide hazard prediction compared with precipitation-only threshold approaches [5]. Their results also highlight a practical issue: the temporal precision of landslide inventories can limit the evaluation of high-frequency warning models.

This observation is important for the proposed framework. A warning system cannot be evaluated more precisely than the event labels allow. If an inventory provides only a calendar date, the experiment should not claim an exact hour of prediction. Therefore, the proposed implementation will define the temporal resolution and event-matching window based on the selected dataset.

C. Temporal and Sequential AI Models

Landslide processes are influenced by antecedent environmental states, making temporal information important. Sequential models such as recurrent neural networks, gated recurrent units, and long short-term memory networks can represent dependencies across observations. Other approaches use rolling statistics and engineered temporal features with conventional machine-learning algorithms.

A sequential model is not automatically better than a static classifier. Its suitability depends on the amount and quality of historical data, the regularity of observations, and the accuracy of event timestamps. A recent review of AI applications in landslide risk assessment noted the growing role of RNN-type approaches in displacement prediction and other temporal tasks [1]. The proposed study therefore treats model selection as an empirical question. A conventional baseline should be included, and a temporal model can be added if the dataset supports it.

D. Multi-Source Monitoring and False Alarms

Early-warning systems must balance sensitivity and reliability. A system that generates a warning for almost every environmental fluctuation may achieve high sensitivity but create too many false alarms. Conversely, a conservative system may avoid false warnings while missing actual events.

Recent work has explicitly considered this issue. A 2026 study proposed an early-warning model using multi-source monitoring data and unsupervised machine learning. The authors identified preset thresholds as a challenge across sites with different geological conditions and reported that combining monitoring sources can reduce false alarms [6]. This provides a strong motivation for evaluating false alarms as a first-class metric rather than treating accuracy as the only outcome.

The proposed framework follows this principle. Risk escalation is not defined as a single high prediction alone. Instead, a configurable temporal rule or learned escalation criterion is evaluated. The criterion should be selected using the validation set and then frozen before testing. This prevents the test data from being used to tune the warning rule.

E. AI, Generalization, and Spatial Heterogeneity

A major challenge in landslide AI is generalization. A model trained in one geographical environment may not perform equally well in another because rainfall regimes, geology, soil properties, terrain, and land use differ. Recent reviews have identified spatial heterogeneity, data quality, overfitting, and model interpretability as important challenges [2], [7].

This motivates the use of location-aware evaluation. If multiple locations are available, the proposed study can compare within-location and cross-location performance. If only one region is available, the limitation should be explicitly reported rather than presenting the model as universally generalizable.

III. Research Gap and Proposed Contribution

A. Problem Definition

Let a location be represented by s and time by t. Let the environmental feature vector be:

$$\mathbf{x}_{s,t} = [x_{1,s,t}, x_{2,s,t}, \ldots, x_{m,s,t}]. \tag{1}$$

The target variable $y_{s,t}$ indicates whether a landslide event occurs within a predefined future prediction window. The prediction model estimates:

$$p_{s,t} = P(y_{s,t+h} = 1 | \mathbf{x}_{s,t}), \tag{2}$$

where h represents the selected prediction horizon.

The key issue is that $p_{s,t}$ is a sequence over time. A warning system should therefore consider:

$$\mathcal{P}_{s,t} = [p_{s,t-k+1}, \ldots, p_{s,t-1}, p_{s,t}]. \tag{3}$$

The proposed framework uses this sequence to determine whether the risk is escalating.

B. Gap Identified

The literature demonstrates substantial activity in landslide susceptibility mapping, AI-based detection, displacement prediction, and rainfall or multi-source early warning. Therefore, the present work does not claim that temporal prediction, soil-moisture integration, dynamic thresholds, or false-alarm analysis are individually new.

Instead, the research gap targeted by this project is methodological integration and evaluation. The proposed framework places future-risk prediction, temporal escalation detection, event-based lead-time measurement, and warning reliability within one reproducible workflow. The contribution is thus defined as an evaluation-oriented framework rather than a claim of a completely new AI algorithm.

C. Research Questions

The study investigates four research questions:

1) RQ1: How accurately can the selected AI model estimate future landslide risk from historical environmental observations?
2) RQ2: Does monitoring the temporal risk trajectory identify meaningful escalation before observed landslide events?
3) RQ3: What lead time is obtained for successfully detected historical events?
4) RQ4: What is the trade-off between early detection and false alarms?

D. Contributions

The intended contributions are:

1) a temporally aligned data-processing pipeline for environmental observations and landslide events;
2) an AI-based future-risk prediction module;
3) a temporal risk-escalation module operating on the predicted risk sequence;
4) an event-based lead-time evaluation protocol;

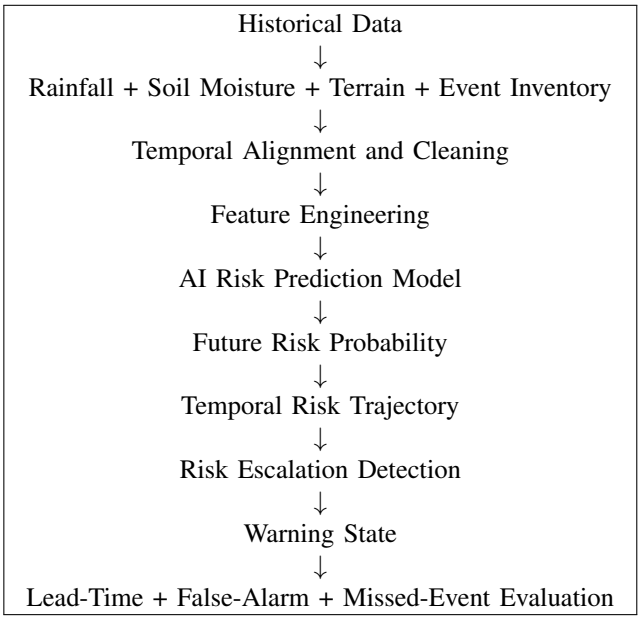


Fig. 1. Conceptual architecture of the proposed framework.

5) an explicit false-alarm and missed-event evaluation procedure; and
6) an experimental framework that reports both conventional predictive metrics and warning-oriented metrics.

IV. Proposed System Architecture

The proposed architecture consists of six principal layers: data acquisition, preprocessing, feature construction, risk prediction, escalation detection, and event-based evaluation.

A. Data Acquisition Layer

The system requires historical observations that can be aligned with landslide events. Candidate variables include rainfall, soil moisture, temperature, slope, elevation, land cover, drainage-related information, and satellite-derived indicators. The final feature set must be determined from the selected dataset.

Rainfall can be represented using daily or sub-daily measurements. Soil moisture can be represented as an absolute value or a normalized anomaly. Terrain variables are generally static over the observation period, although dynamic surface-change variables can be added if available.

B. Temporal Alignment

Temporal alignment is essential because environmental data and landslide inventories often use different timestamps. The framework maps each environmental observation to the corresponding location and event window.

If a landslide is recorded on day T_e, an antecedent window can be defined as:

$$W_e = [T_e - w, T_e - 1], \tag{4}$$

where w is the selected historical window length.

TABLE I
Illustrative Warning States

State	Interpretation	System Output
Green	Low / stable	Monitor
Yellow	Increasing	Observe
Orange	Sustained escalation	Prepare
Red	High / critical	Early warning

The prediction target can then be defined for a future horizon h. For example, the system may learn whether an event occurs within the next h observations. The actual value of h must be determined by the dataset and research objective; it should not be chosen simply to produce a favorable result.

C. Risk Prediction Layer

The prediction layer converts the environmental state into a probability or risk score. A probability-based output is preferable because it provides a continuous value that can later be analysed as a trajectory.

For a classifier:

$$p_t = f_\theta(\mathbf{x}_t), \tag{5}$$

where θ denotes the model parameters.

The risk score can be visualized over time. A hypothetical sequence is:

$$0.20, 0.23, 0.28, 0.35, 0.45, 0.60, 0.68, 0.75, 0.82, 0.90. \tag{6}$$

This sequence is illustrative only and is not a reported experimental result.

D. Risk Escalation Layer

The escalation layer evaluates the trajectory instead of a single observation. A simple baseline rule can use the slope of the risk sequence:

$$S_t = \frac{p_t - p_{t-k}}{k}. \tag{7}$$

A positive value indicates increasing predicted risk over the selected interval. A warning condition can then be expressed as:

$$W_t = \begin{cases} 1, & S_t > \tau_s \text{ and } p_t > \tau_p, \\ 0, & \text{otherwise}, \end{cases} \tag{8}$$

where τ_s is an escalation-rate threshold and τ_p is a risk threshold.

The important methodological point is that these parameters should be learned or selected on the validation data. They should not be manually tuned on the final test set.

TABLE II
REQUIRED DATASET COMPONENTS

Component	Description
Rainfall	Time-varying precipitation measurements
Soil moisture	Ground moisture state or proxy
Terrain	Elevation, slope, aspect or related variables
Location	Geographic identifier or coordinates
Event inventory	Landslide occurrence and timestamp/date
Optional remote sensing	Surface deformation or land-cover change

E. Warning States

For practical visualization, the continuous risk can be converted into warning states. A four-level representation can be used:

The actual state boundaries should be derived from validation experiments and domain requirements. They should not be presented as universally valid thresholds.

F. Lead-Time Layer

Once a warning is generated, it is matched with an observed event. Let T_w be the first valid warning time and T_e be the event time:

$$L = T_e - T_w. \tag{9}$$

If the event occurs after the warning, $L > 0$. If an event occurs without a valid preceding warning, it is counted as a missed event according to the evaluation protocol.

For example, if a warning is generated on June 25 and the event is recorded on June 30, the lead time is five days. This example is used only to explain the metric.

V. DATASET DESIGN AND DATA PREPROCESSING

A. Dataset Requirements

The dataset should contain three categories of information: environmental variables, spatial information, and historical event labels.

The selected dataset should be sufficiently long to contain both event and non-event periods. A small dataset with only a few landslides may be inadequate for reliable event-based evaluation.

B. Missing Values

Environmental datasets can contain missing observations because of sensor failure, cloud contamination, data transmission problems, or incomplete historical records. Missing values should be handled without introducing information from the future.

Possible strategies include forward filling for suitable slowly varying variables, interpolation when justified, or removal of observations with excessive missingness. The strategy must be documented for reproducibility.

C. Outlier Handling

Extreme rainfall values or soil-moisture observations should not automatically be removed because extreme environmental conditions may be precisely the conditions associated with landslides. Therefore, outlier analysis should distinguish between measurement errors and physically meaningful extremes.

A robust approach is to inspect distributional characteristics and compare suspicious values with source metadata. Values confirmed as valid extremes should remain in the dataset.

D. Normalization

For algorithms sensitive to feature scale, continuous variables can be standardized:

$$z = \frac{x - \mu}{\sigma}, \tag{10}$$

where μ is the training-set mean and σ is the training-set standard deviation.

The mean and standard deviation must be computed using the training data only. Applying statistics computed from the complete dataset would introduce information leakage.

E. Temporal Features

The framework can construct cumulative rainfall features:

$$R_{k,t} = \sum_{i=0}^{k-1} R_{t-i}. \tag{11}$$

It can also calculate moisture change:

$$\Delta M_{k,t} = M_t - M_{t-k}. \tag{12}$$

Additional features can include rolling mean, rolling maximum, rolling standard deviation, rainfall intensity, antecedent rainfall, and interaction terms. Feature windows should be selected through validation rather than by inspecting the final test period.

F. Class Imbalance

Landslide events are generally less frequent than non-event observations. This creates class imbalance. Accuracy can become misleading if the model predicts the majority class most of the time.

Possible solutions include class-weighted learning, controlled sampling, or threshold optimization. Oversampling must be performed carefully in temporal data because randomly mixing observations can cause leakage between training and testing periods.

VI. MODEL DEVELOPMENT

A. Baseline Models

A strong experimental study should begin with simple baselines. Logistic regression can provide an interpretable reference. Random forest can provide a nonlinear tree-based baseline. A gradient boosting model can provide another strong tabular-data reference.

The objective is not to assume that a complex deep-learning model is superior. Instead, the final model should be selected based on validation performance, stability, computational requirements, and suitability to the temporal structure.

B. Temporal Model

If the dataset contains sufficiently dense and long sequences, a temporal neural network can be investigated. An input sequence can be represented as:

$$\mathbf{X}_t = [\mathbf{x}_{t-k+1}, \ldots, \mathbf{x}_t]. \tag{13}$$

The temporal model produces:

$$p_t = f_\theta(\mathbf{X}_t). \tag{14}$$

A recurrent or attention-based architecture can be evaluated against the tabular baseline. However, the model should not be selected solely because it is more complex.

C. Model Selection

The validation set should be used to compare candidate models. The primary selection metric should reflect the research objective. Because the event class may be imbalanced, F1-score, recall, precision-recall AUC, and ROC-AUC can be considered together.

The final test set must remain untouched during model selection.

D. Explainability

AI-based warning systems should provide interpretable evidence when possible. Feature importance methods can be used to identify variables associated with predictions. For tree-based models, permutation importance or model-specific importance measures can be used. For more complex models, SHAP-like approaches may be considered if computationally feasible.

Explainability should be used to describe model behaviour, not to claim physical causation. A feature that is important to the model is not automatically a proven physical trigger.

VII. Risk Escalation and Early Warning Algorithm

The proposed warning algorithm contains three stages.

A. Stage 1: Future Risk Estimation

At every valid observation time, the model estimates the future landslide probability:

$$p_t = f_\theta(\mathbf{x}_t). \tag{15}$$

The value is stored in a chronological risk series.

TABLE III
Illustrative Temporal Split

Subset	Time Period
Training	Earliest historical period
Validation	Subsequent period
Testing	Latest unseen period

B. Stage 2: Escalation Detection

The system evaluates whether risk has increased consistently over a selected window. A generic escalation score can be defined as:

$$E_t = \alpha(p_t - p_{t-1}) + \beta(p_t - p_{t-k}) + \gamma C_t, \tag{16}$$

where C_t represents the number or proportion of positive changes within the window and α, β, γ are validation parameters.

The exact formula can be simplified if the selected dataset does not support a more complex approach. The principle is to distinguish a persistent increase from a single noisy fluctuation.

C. Stage 3: Warning Generation

A warning is produced when both predicted risk and escalation satisfy validation criteria. A persistence condition can be included so that one isolated spike does not immediately trigger a warning.

For example:

$$W_t = 1 \tag{17}$$

if the risk exceeds the selected level for at least q consecutive observations and the escalation score is positive. The values of q and thresholds must be selected using training/validation data.

D. Event Matching

A warning is considered successful if an observed landslide occurs within the predefined event window after the first valid warning. If multiple warnings occur before one event, only the first qualifying warning should normally be counted for lead-time analysis to avoid artificially inflating the number of warnings.

This event-matching rule must be fixed before final testing.

VIII. Experimental Design

A. Time-Aware Data Split

Random train-test splitting is inappropriate when future information can leak into the training set. The proposed study therefore uses chronological splitting.

A possible design is:

The actual dates depend on the selected dataset.

B. Evaluation Metrics

For binary prediction, let TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives.

Accuracy is:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}. \tag{18}$$

Precision is:

$$Precision = \frac{TP}{TP + FP}. \tag{19}$$

Recall is:

$$Recall = \frac{TP}{TP + FN}. \tag{20}$$

The F1-score is:

$$F1 = 2\frac{Precision \cdot Recall}{Precision + Recall}. \tag{21}$$

For warning applications, event-level detection should also be reported:

$$DetectionRate = \frac{N_{detected}}{N_{events}}. \tag{22}$$

Average lead time is:

$$\overline{L} = \frac{1}{N_{detected}} \sum_{i=1}^{N_{detected}} L_i. \tag{23}$$

A warning-oriented false-alarm rate can be defined according to the event-matching protocol. The exact denominator should be stated clearly because multiple definitions are possible.

C. Ablation Experiments

To understand the contribution of different variables, the study can perform ablation experiments:

1) rainfall only;
2) rainfall + soil moisture;
3) rainfall + terrain;
4) rainfall + soil moisture + terrain;
5) full available feature set.

This analysis helps determine whether multi-source data provide measurable benefit.

D. Lead-Time Sensitivity

The escalation window affects warning timing. A short window may react quickly but can be sensitive to noise. A long window may produce more stable warnings but reduce lead time.

Therefore, several validation-window settings can be compared. The selected setting should be fixed before testing.

TABLE IV
MODEL PERFORMANCE ON THE UNSEEN TEST SET

Model	Accuracy	Precision	Recall	F1	AUC
Logistic Regression	–	–	–	–	–
Random Forest	–	–	–	–	–
Gradient Boosting	–	–	–	–	–
Temporal Model	–	–	–	–	–
Proposed Selected Model	–	–	–	–	–

PLACEHOLDER FOR ACTUAL RISK-TREND FIGURE
X-axis: Date / Time
Y-axis: Predicted Landslide Risk
Mark: First Escalation Warning
Mark: Actual Landslide Event

Fig. 2. Representative temporal risk trajectory. Replace with an experimentally generated plot.

IX. RESULTS AND DISCUSSION

A. Important Note on Results

The numerical results in this section must be filled only after the proposed system is implemented and evaluated on the selected dataset. This paper intentionally does not invent accuracy, F1-score, lead-time, or false-alarm values.

B. Prediction Results

The final paper should report the performance of each baseline and the proposed model.

The discussion should not simply identify the highest accuracy. It should explain whether improvements are consistent across recall, precision, and event-level detection. If the dataset is highly imbalanced, recall and precision should receive particular attention.

C. Risk Trajectory

The paper should include representative risk trajectories for successfully detected events.

The figure should show whether risk increases before the event and whether the warning occurs early enough to be operationally meaningful. It is important to include examples of both successful and unsuccessful warnings.

D. Event-Based Lead Time

The final discussion should report the distribution rather than only the mean. For example, the median, minimum, maximum, and interquartile range can be useful because one very large lead time may otherwise distort the average.

E. False Alarms

False alarms should be analysed by examining the environmental conditions that preceded them. A false warning may occur because rainfall or soil moisture temporarily resembles a historical triggering pattern without resulting in failure.

The analysis should distinguish between model error and uncertain event labels. If the inventory is incomplete, an

TABLE V
EVENT-BASED WARNING EVALUATION

Event	Warning Date	Event Date	Lead Time	Outcome
E1	–	–	–	–
E2	–	–	–	–
E3	–	–	–	–
E4	–	–	–	–
E5	–	–	–	–

TABLE VI
ABLATION STUDY TEMPLATE

Feature Set	F1	AUC	Detection Rate
Rainfall only	–	–	–
Rainfall + Soil Moisture	–	–	–
Rainfall + Terrain	–	–	–
All Available Features	–	–	–

apparent false alarm may correspond to an unrecorded event. Therefore, conclusions about false alarms should acknowledge inventory completeness.

F. Missed Events

A missed event occurs when an observed landslide is not preceded by a valid warning according to the predefined evaluation window. Missed events are important because a warning system with high sensitivity but several missed major events may still be unsuitable.

For each missed event, the paper should inspect whether:

1) environmental variables were unavailable;
2) the event occurred outside the model's prediction horizon;
3) the model underestimated the risk;
4) the event label had insufficient temporal precision; or
5) the event was associated with factors not represented in the feature set.

G. Ablation Results

The ablation experiment should demonstrate whether each data source contributes useful information.

If soil moisture improves performance, the discussion should relate the result to the physical interpretation of antecedent wetness. If the feature does not improve performance, that result should also be reported rather than omitted.

H. Comparison With Existing Approaches

The comparison should focus on methodological characteristics rather than claiming superiority without matched experiments.

This comparison should be updated with the exact papers selected for the final literature review. The proposed framework should not be described as universally better; its contribution is the integration of risk trajectory analysis and event-oriented warning evaluation.

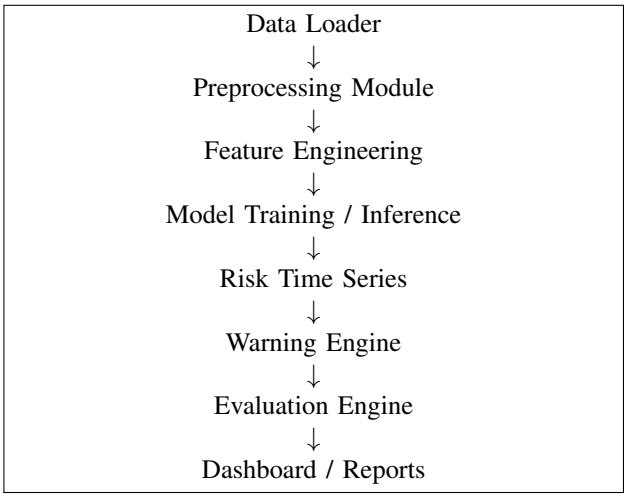


Fig. 3. Prototype software architecture.

X. IMPLEMENTATION AND SYSTEM PROTOTYPE

A. Software Architecture

The prototype can be implemented as a modular pipeline. A data module loads environmental observations and event labels. A preprocessing module performs cleaning and temporal alignment. A model module performs training and inference. A warning module converts predictions into escalation states. An evaluation module calculates event-level metrics. A visualization layer displays risk trajectories and warnings.

A possible architecture is:

B. Dashboard Design

The final prototype can provide:

- current predicted risk;
- historical risk trajectory;
- warning state;
- detected escalation time;
- event date when available;
- estimated lead time for historical validation;
- major contributing variables; and
- warning history.

The dashboard is a visualization and decision-support layer. It should not be presented as an autonomous emergency decision-maker. Real-world evacuation or public-safety decisions require appropriate authorities and validated operational systems.

C. Data Flow

At inference time, the system receives new environmental observations. The same preprocessing transformations used during training are applied. The trained model generates a risk estimate. The warning engine updates the risk trajectory and checks the escalation rule.

This separation is important because training-time transformations must not be recalculated using future test information.

TABLE VII
COMPARISON OF RESEARCH CHARACTERISTICS

Approach	Spatial Susceptibility	Temporal Prediction	Event Lead Time	False-Alarm Evaluation
Susceptibility Mapping	Yes	Usually limited	Not central	Usually not central
Rainfall Threshold	Limited	Yes	Possible	Often threshold-dependent
AI Risk Prediction	Yes/No	Yes	Can be evaluated	Can be evaluated
Multi-source Monitoring	Site dependent	Yes	Possible	Important in warning studies
Proposed Framework	Yes/optional	Yes	Explicitly evaluated	Explicitly evaluated

XI. ROBUSTNESS, UNCERTAINTY, AND RELIABILITY

A. Prediction Uncertainty

A probability estimate is not equivalent to certainty. Two predictions of 0.80 may have different reliability depending on the calibration of the model and the distribution of the test data.

Calibration can therefore be evaluated using reliability diagrams or calibration error measures. If probabilities are poorly calibrated, the warning thresholds should be interpreted as model scores rather than literal probabilities.

B. Temporal Stability

The model should be evaluated across different time periods. Performance may vary during unusually wet or dry periods. Reporting only the overall average can hide temporal instability.

A rolling or blocked evaluation can reveal whether the model remains useful across seasons.

C. Spatial Stability

If multiple locations are available, the study should compare performance across locations. A location-adaptive design can be evaluated by training and testing within the same region and by testing on a geographically distinct region when sufficient data are available.

The second experiment is more difficult but provides stronger evidence about generalization.

D. Sensitivity to Missing Data

A practical warning system must tolerate incomplete observations. The experiment can simulate controlled missingness and evaluate how performance changes. This is particularly important for remote sensing and sensor-based data.

XII. ETHICAL AND OPERATIONAL CONSIDERATIONS

The proposed system is a research prototype and should not be represented as a certified emergency-warning system. A false warning can cause unnecessary disruption, while a missed event can have serious consequences. Therefore, model performance must be communicated transparently.

The study should report the limitations of the landslide inventory, the uncertainty of event timestamps, the geographical scope, and the variables used. It should also avoid implying that a model prediction guarantees that a landslide will occur.

For real-world deployment, warnings would require validation with domain experts, local geological knowledge, communication infrastructure, and an appropriate authority-approved emergency protocol.

XIII. LIMITATIONS

Several limitations may affect the proposed study.

First, the quality of the historical landslide inventory directly affects model learning and evaluation. Missing events or inaccurate timestamps can produce noisy labels.

Second, environmental datasets may have different spatial and temporal resolutions. Aggregating them to a common grid or time interval can introduce uncertainty.

Third, the selected model may learn correlations rather than physical causation. Feature importance should therefore not be interpreted as proof of a causal mechanism.

Fourth, a single-region dataset may limit generalization. Results should be described as valid for the studied domain unless cross-region testing is performed.

Fifth, the lead-time value depends on the definition of a valid warning and the temporal precision of event labels. A warning five days before an event recorded only by date is not equivalent to a five-day physically exact prediction.

Sixth, the escalation rule introduces design choices. If these choices are tuned excessively on the validation data, the system may still overfit. Therefore, the final test set must remain untouched until all design decisions are frozen.

XIV. FUTURE WORK

Future work can extend the framework in several directions.

First, satellite-derived deformation observations can be incorporated when appropriate data are available. Surface displacement can provide complementary information to rainfall and soil moisture.

Second, uncertainty-aware models can be investigated so that the system provides both a risk estimate and an uncertainty interval.

Third, transfer learning or domain adaptation can be explored to improve performance across different geographical regions.

Fourth, explainable AI can be integrated to show which environmental variables contribute most strongly to each warning.

Fifth, the warning engine can be extended from a binary warning to a multi-level state model that considers persistence

and hysteresis. Hysteresis can reduce rapid switching between warning states when predictions fluctuate around a threshold.

Sixth, a more rigorous cost-sensitive evaluation can be developed. False alarms and missed events do not have equal operational consequences, so future research can investigate cost-weighted decision criteria.

Seventh, near-real-time data pipelines can be studied. Such a system would require reliable data ingestion, quality control, model inference, and monitoring infrastructure.

XV. Conclusion

This paper proposed an AI-based location-adaptive landslide early-warning framework that combines historical environmental learning, future landslide-risk prediction, temporal risk-escalation detection, and event-based warning evaluation.

The central idea is to move beyond interpreting a landslide-risk value as an isolated output. The system maintains a temporal sequence of predicted risk and investigates whether the risk is persistently increasing before an observed event. When an escalation condition is satisfied, an early-warning state is generated. The warning is subsequently evaluated using lead time, detection rate, false alarms, and missed events.

The framework does not claim that the concepts of AI landslide prediction, temporal analysis, soil-moisture integration, or false-alarm evaluation are individually new. Recent research already demonstrates substantial activity in these areas. The intended research contribution is a reproducible integration and evaluation framework that explicitly connects future-risk prediction with temporal warning performance.

The final experimental study must use a validated historical dataset and report actual results without fabricated values. Model performance should be compared with appropriate baselines, and the test period must remain unseen during model selection. Event-level analysis should report not only average lead time but also the distribution of lead times and the number of false and missed warnings.

The proposed framework can serve as a foundation for a student research prototype and can be extended with additional remote-sensing, deformation, hydrological, or location-specific information. Its ultimate value will depend on the quality of the dataset, the validity of event labels, and the robustness of the experimental evaluation.

Acknowledgment

The authors would like to thank the Department of Computer Science and Engineering and the institution for providing academic guidance and computational resources for this research.

References

[1] Y. Wang et al., "Application of artificial intelligence in three aspects of landslide risk assessment: A comprehensive review," *Rock Mechanics Bulletin*, vol. 3, no. 4, 100144, 2024, doi: 10.1016/j.rockmb.2024.100144.

[2] Z. Lu, G. Liu, Z. Song, K. Sun, M. Li, Y. Chen, X. Zhao, and W. Zhang, "Advancements in technologies and methodologies of machine learning in landslide susceptibility research: Current trends and future directions," *Applied Sciences*, vol. 14, no. 21, 9639, 2024, doi: 10.3390/app14219639.

[3] Y. Han and S. J. Semnani, "Important considerations in machine learning-based landslide susceptibility assessment under future climate conditions," *Acta Geotechnica*, vol. 20, pp. 475–500, 2025, doi: 10.1007/s11440-024-02363-3.

[4] S. M. Kassa, "A systematic review of machine learning based landslide susceptibility mapping," *Journal of Road and Traffic Engineering*, vol. 70, no. 2, pp. 23–30, 2024, doi: 10.31075/PIS.70.02.03.

[5] T. Halter, P. Lehmann, A. Bast, J. Aaron, M. Stähli, et al., "In situ soil moisture data improve precipitation-based shallow landslide early warning through innovative machine learning methods," *Landslides*, vol. 22, pp. 3599–3614, 2025, doi: 10.1007/s10346-025-02599-4.

[6] "Landslide early warning model based on multi-source monitoring data and unsupervised machine learning," *Engineering Applications of Artificial Intelligence*, vol. 164, 113156, 2026, doi: 10.1016/j.engappai.2025.113156.

[7] M. Ehsan, M. T. Anees, A. F. B. A. Bakar, et al., "A review of geological and triggering factors influencing landslide susceptibility: artificial intelligence-based trends in mapping and prediction," *International Journal of Environmental Science and Technology*, vol. 22, pp. 17347–17382, 2025.

[8] "Modelling landslide susceptibility prediction: A review and construction of semi-supervised imbalanced theory," *Earth-Science Reviews*, vol. 250, 104700, 2024, doi: 10.1016/j.earscirev.2024.104700.